

BACKEND SCHEMA DOCUMENTATION

Carbon Horizon

Learn Your Impact. Shape the Future.

Version: 2.0

1. Database Overview

Purpose

The Carbon Horizon database stores:

- User accounts
- Carbon assessments
- Emission inputs
- Simulations
- Forecasts
- Goals
- Habit tracking
- AI conversations
- Carbon factor references

The schema is designed to:

- Support MVP requirements
 - Maintain scientific traceability
 - Enable future scalability
 - Preserve historical assessment data
 - Support AI-powered recommendations
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2. Database Technology

Development Database: SQLite

Production Database: PostgreSQL 16+

ORM: SQLAlchemy

Migration Tool: Alembic

Primary Key Strategy: UUID

Timestamp Strategy: UTC Timestamps

Soft Delete Strategy: deleted_at field

3. Entity Relationship Overview

User |—— Carbon Assessments | |—— Emission Inputs |—— Simulations |—— Forecasts | |——
Forecast Points |—— Goals |—— Habits |—— AI Conversations |—— User Preferences

Carbon Factors |—— Assessment Calculations

4. Users Table

Table Name

users

Purpose

Stores authentication and profile information.

Fields

id UUID PRIMARY KEY

full_name VARCHAR(100) NOT NULL

email VARCHAR(255) UNIQUE NOT NULL

password_hash VARCHAR(255) NOT NULL

age_group ENUM (child, student, adult, senior)

lifestyle_type ENUM (student, professional, homemaker, retired)

city VARCHAR(100)

country VARCHAR(100)

email_verified BOOLEAN DEFAULT FALSE

last_login TIMESTAMP

created_at TIMESTAMP

updated_at TIMESTAMP

deleted_at TIMESTAMP

5. User Preferences Table

Table Name

user_preferences

Purpose

Stores user-specific settings.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

theme ENUM (light, dark, system)

language VARCHAR(20)

notifications_enabled BOOLEAN

measurement_unit ENUM (metric, imperial)

created_at TIMESTAMP

updated_at TIMESTAMP

6. Carbon Assessments Table

Table Name

carbon_assessments

Purpose

Stores carbon footprint calculations.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

transport_emission FLOAT

energy_emission FLOAT

food_emission FLOAT

waste_emission FLOAT

total_emission FLOAT

carbon_score INTEGER

calculation_version VARCHAR(20)

factor_version VARCHAR(20)

assessment_period ENUM (daily, monthly, annual)

created_at TIMESTAMP

updated_at TIMESTAMP

7. Emission Inputs Table

Table Name

emission_inputs

Purpose

Stores raw assessment data.

Fields

id UUID PRIMARY KEY

assessment_id UUID FOREIGN KEY → carbon_assessments.id

transport_mode ENUM (car, motorcycle, bus, train, flight, bicycle)

distance_km FLOAT

electricity_kwh FLOAT

ac_hours FLOAT

lpg_usage FLOAT

solar_usage BOOLEAN

diet_type ENUM (vegetarian, mixed, non_vegetarian)

recycling_score INTEGER

plastic_usage_score INTEGER

household_size INTEGER

created_at TIMESTAMP

8. Sustainability Goals Table

Table Name

goals

Purpose

Tracks user sustainability objectives.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

goal_name VARCHAR(255)

goal_description TEXT

target_reduction_percentage FLOAT

target_emission_value FLOAT

current_progress FLOAT

target_date DATE

status ENUM (active, completed, expired)

created_at TIMESTAMP

updated_at TIMESTAMP

9. Simulations Table

Table Name

simulations

Purpose

Stores simulation scenarios.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

scenario_name VARCHAR(255)

scenario_description TEXT

current_emission FLOAT

projected_emission FLOAT

reduction_percentage FLOAT

estimated_carbon_saved FLOAT

simulation_data JSONB

created_at TIMESTAMP

updated_at TIMESTAMP

10. Forecasts Table

Table Name

forecasts

Purpose

Stores forecast sessions.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

forecast_type ENUM (current_path, recommended_path, custom_path)

created_at TIMESTAMP

updated_at TIMESTAMP

11. Forecast Points Table

Table Name

forecast_points

Purpose

Stores forecast values dynamically.

Why This Table Exists

Instead of storing:

predicted_3_month predicted_6_month predicted_12_month

we store unlimited forecast points.

This supports:

- 3 Months
- 6 Months
- 12 Months
- 24 Months
- Future Expansion

Fields

id UUID PRIMARY KEY

forecast_id UUID FOREIGN KEY → forecasts.id

month_offset INTEGER

predicted_emission FLOAT

created_at TIMESTAMP

12. Habit Tracking Table

Table Name

habits

Purpose

Tracks sustainability habits.

Fields

id UUID PRIMARY KEY

user_id UUID FOREIGN KEY → users.id

habit_type ENUM (public_transport, recycling, save_electricity, water_conservation, plastic_reduction)

completed BOOLEAN

carbon_saved FLOAT

activity_date DATE

notes TEXT

created_at TIMESTAMP

13. Habit Definitions Table

Table Name

habit_definitions

Purpose

Stores standardized habit-to-carbon conversion factors.

Example

Public Transport Used

= 1.2 kg CO₂e saved

Fields

id UUID PRIMARY KEY

habit_type VARCHAR(100)

carbon_saving_factor FLOAT

unit VARCHAR(50)

description TEXT

created_at TIMESTAMP

updated_at TIMESTAMP

14. AI Conversations Table

Table Name

ai_conversations

Purpose

Stores AI chat history.

Fields

id UUID PRIMARY KEY

conversation_id UUID

user_id UUID FOREIGN KEY → users.id

role ENUM (user, assistant)

message TEXT

created_at TIMESTAMP

15. Carbon Factors Table

Table Name

carbon_factors

Purpose

Stores official emission factors.

Fields

id UUID PRIMARY KEY

category VARCHAR(100)

sub_category VARCHAR(100)

factor_value FLOAT

unit VARCHAR(50)

source_reference TEXT

source_name VARCHAR(255)

version VARCHAR(20)

effective_date DATE

updated_at TIMESTAMP

16. Assessment History View

View Name

assessment_history_view

Purpose:

Provide dashboard-ready assessment history.

Contains:

- Assessment Date
 - Total Emission
 - Carbon Score
 - Trend Delta
-

17. Dashboard Summary View

View Name

dashboard_summary_view

Purpose:

Fast dashboard rendering.

Contains:

- Latest Assessment
 - Active Goals
 - Current Streak
 - Forecast Summary
 - Total Carbon Saved
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18. API Response Models

Carbon Assessment Response

```
{ "assessment_id": "uuid", "total_emission": 2.8, "transport": 1.3, "energy": 0.8, "food": 0.5, "waste": 0.2, "carbon_score": 78 }
```

Simulation Response

```
{ "scenario_name": "Switch To Public Transport", "current": 3.2, "projected": 2.4, "saved": 0.8, "reduction_percentage": 25 }
```

Forecast Response

```
{ "forecast_type": "recommended_path", "forecast_points": [ { "month": 3, "emission": 3.0 }, { "month": 6, "emission": 2.7 }, { "month": 12, "emission": 2.3 } ] }
```

19. Relationships

users 1 → N carbon_assessments

users 1 → N goals

users 1 → N habits

users 1 → N simulations

users 1 → N forecasts

users 1 → N ai_conversations

users 1 → 1 user_preferences

carbon_assessments 1 → 1 emission_inputs

forecasts 1 → N forecast_points

habit_definitions 1 → N habits

20. Indexing Strategy

Indexes:

users.email

carbon_assessments.user_id

goals.user_id

habits.user_id

habits.activity_date

simulations.user_id

forecasts.user_id

forecast_points.forecast_id

ai_conversations.conversation_id

carbon_factors.category

carbon_factors.sub_category

Purpose:

- Faster dashboard loading
 - Faster historical lookups
 - Improved forecast retrieval
 - Faster AI conversation history access
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21. Data Retention Policy

Assessment Data: Retained indefinitely

Simulation Data: Retained indefinitely

Habit Data: Retained indefinitely

AI Conversations: Retained for 24 months

Soft Deleted Accounts: Retained for 90 days

Audit Logs: Retained for 12 months

22. Future Tables (Version 2+)

bill_uploads

ocr_results

community_challenges

leaderboards

carbon_rewards

carbon_budgets

organization_accounts

school_programs

iot_devices

energy_meter_readings

These tables are intentionally excluded from MVP to maintain development focus and competition readiness.

23. Schema Success Criteria

The database schema shall be considered successful if:

- All MVP features are supported
- Historical assessments remain traceable
- Carbon factors are version-controlled
- Forecasts support unlimited time horizons
- Goal tracking operates correctly
- Dashboard queries execute efficiently
- Future feature expansion requires minimal schema changes